BRIGHTNESS & NITS CHEAT SHEET

LED display brightness specification for UAE/GCC environments | BRIGHTLUXX

Environment	Ambient Lux	Min Nits	Recommended	Notes
Dark room / Cinema	0-50 lux	100-300	200-500	Low brightness, high contrast priority
Meeting room	300-500 lux	300-600	600-800	Standard office lighting
Hotel lobby	500-1000 lux	600-1000	800-1200	Mixed lighting, feature wall
Retail store	800-1500 lux	1000-1500	1200-1800	Bright retail environments
Mall atrium	1000-2000 lux	1500-2000	2000-2500	Skylights, mixed natural light
Semi-outdoor canopy	2000-10000 lux	2500-3500	3500-5000	UAE covered walkways
Outdoor (shade)	10k-30k lux	3500-5500	5500-7000	North-facing, shaded
Outdoor (direct sun)	50k-100k lux	5500-7000	7000-10000	South/west-facing Dubai

UAE-SPECIFIC GUIDANCE

Dubai direct sun reaches 100,000+ lux. Specify 7,000-10,000 nits for south/west-facing outdoor. Add 20-30% headroom for thermal derating at 45C+. East-facing is easier — morning sun is less intense.

For semi-outdoor (parking canopies, covered walkways, pool decks): IP54 minimum + 3,500-5,000 nits.

POWER IMPACT

Higher brightness = higher power consumption and heat generation. Outdoor 10,000-nit displays draw 600-800W/sqm vs indoor 250W/sqm. Ensure HVAC compensates for heat output.